

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Statement 1 | Every element of a group generates a cyclic subgroup of the group. Statement 2 | The symmetric group S_{10} has 10 elements.' is 'True, True'.
2. True or False: The correct answer to 'Statement 1 | If H is a subgroup of a group G and a belongs to G , then $aH = Ha$. Statement 2 | If H is normal of G and a belongs to G ,...' is 'False, True'.
3. True or False: The correct answer to 'Find the product of the given polynomials in the given polynomial ring. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$.' is ' $x^2 + 1$ '.
4. True or False: The correct answer to 'Let V and W be 4-dimensional subspaces of a 7-dimensional vector space X . Which of the following CANNOT be the dimension of the subspace...' is '2'.
5. True or False: The correct answer to 'Statement 1 | If a and b are elements of finite order in an Abelian group, then $|ab|$ is the lcm $(|a|, |b|)$. Statement 2 | If g is a group...' is 'True, True'.
6. True or False: The correct answer to 'Compute the product in the given ring. $(20)(-8)$ in $\mathbb{Z}_{26}$ ' is '22'.
7. True or False: The correct answer to 'Statement 1 | Every ideal in a ring is a subring of the ring. Statement 2 | Every subring of every ring is an ideal of the ring.' is 'True, False'.
8. True or False: The correct answer to 'Statement 1 | A factor group of a non-Abelian group is non-Abelian. Statement 2 | If K is a normal subgroup of H and H is a normal...' is 'False, False'.
9. True or False: The correct answer to 'Statement 1 | $\mathbb{R}$ is a splitting field of some polynomial over $\mathbb{Q}$. Statement 2 | There is a field with 60 elements.' is 'True, True'.
10. True or False: The correct answer to 'How many positive numbers x satisfy the equation $\cos(97x) = x$?' is '31'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Find the integer closest in value to

$$\frac{1}{\sqrt[4]{5^4 + 1} - \sqrt[4]{5^4 - 1}}.$$

2. A function f is defined by $f(z) = i\bar{z}$, where $i^2 = -1$ and $\bar{z}$ is the complex conjugate of z . How many values of z satisfy both $|z| = 5$ and $f(z) = z$?

End of Paper